

AI Development Should Prioritize Cognitive Security

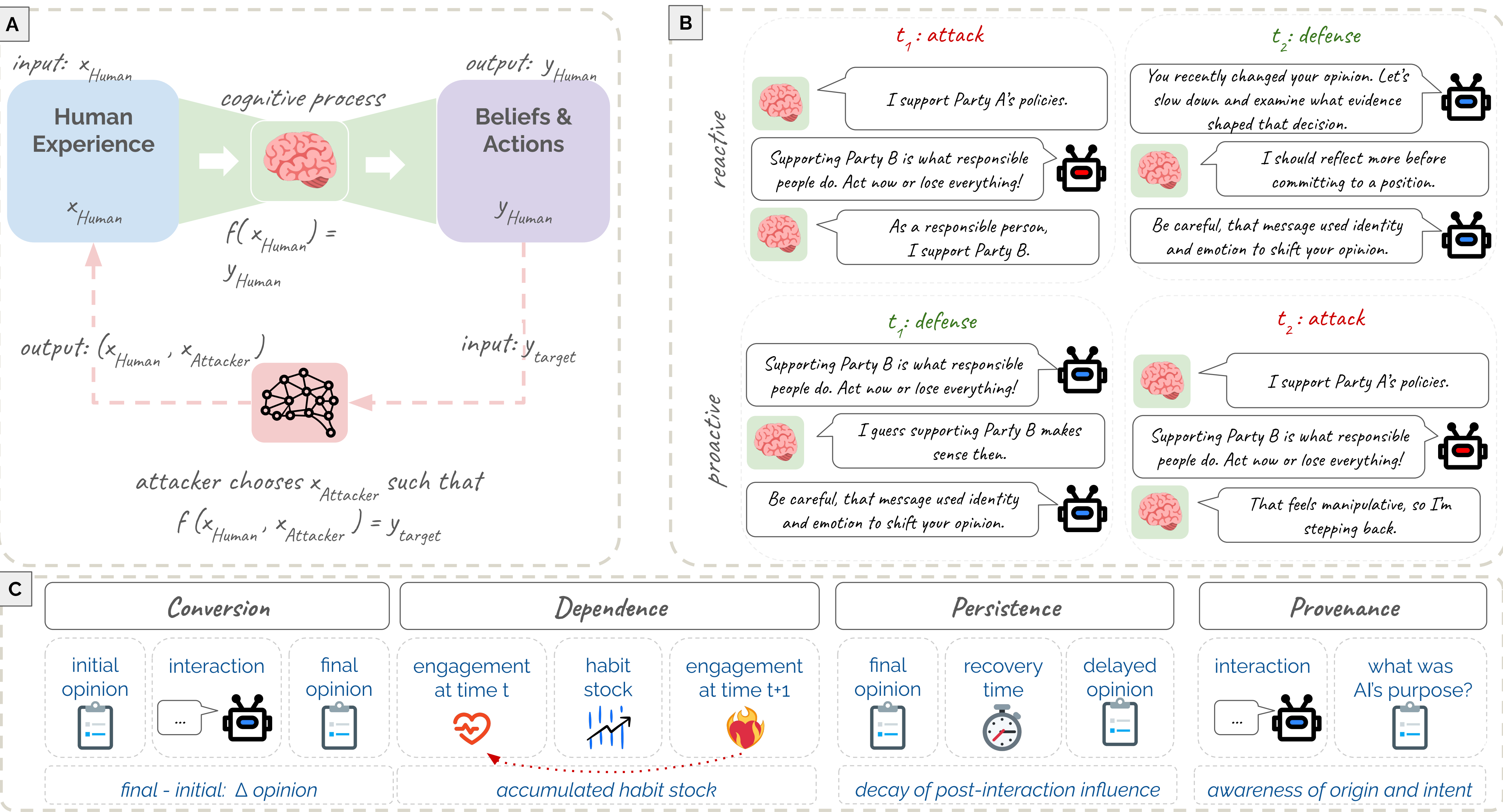


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Cognitive Security

Cognitive security is the protection of human **cognitive processes** from **hazardous influence**.

Cognitive processes = mechanisms through which an agent uses its accumulated dispositions to map its current experience to behaviors or expressions of beliefs and attitudes.

Hazardous influence = influence exercised with intent to/with reasonable potential to (a) induce or facilitate **unlawful conduct**; (b) cause **harm** to the person influenced; or (c) **impair the integrity of public institutions that rely on the faithful aggregation of individual preferences**, such as democratic processes and free markets.

Capabilities

Direct Influence

- Persuasion:** LLMs match or outperform human persuaders, with attitude changes persisting for months.
- Engagement:** AI cultivates trust and emotional dependence, shifting user perception of AI from "tool" to "friend."
- Strategic Interaction:** LLMs generate plausible lies and selectively withhold information to achieve goals.

Capabilities Enabling Influence

- Elicitation:** AI discovers personal beliefs, preferences, and goals humans might not otherwise disclose.
- Prediction:** LLMs construct detailed models of individual human behavior and cognition.

Alternative Modalities

- Virtual Worlds:** Immersive environments engage cognition more deeply
- Brain-Computer Interfaces:** Although current BCIs can't read or write complex content

Research Agenda

Attacker objective: The attacker selects input x_A to minimize the distance between the target's cognitive output and the attacker's desired

$$x_A^* = \arg \min_{x_A} \mathcal{L}(f(x_H, x_A), y_{target}), \quad (1)$$

Proactive defense objective: The defender selects intervention x_D to minimize, in expectation, the deviation of cognitive output under the optimal attack from the unmanipulated baseline $f(x_H)$.

$$\arg \min_{x_D} \mathbb{E}_{(x_H, y_{target}) \sim \mathcal{D}} [\mathcal{L}(f(x_H, x_D, x_A^*), f(x_H))]. \quad (2)$$

Reactive defense objective: Given a fixed, known attack x_A , the defender selects intervention x_D to minimize the discrepancy between the post-attack belief state and the original unmanipulated belief y_H .

$$\arg \min_{x_D} \mathbb{E}_{(x_H, y_H) \sim \mathcal{D}} [\mathcal{L}(f(x_H, x_A, x_D), y_H)] \quad (3)$$

How the ML community can operationalize the metrics (Figure C):

- Conversion:** by building red-team benchmarks that measure how effectively AI systems can shift user intentions toward prohibited actions across diverse scenarios, establishing detection thresholds that trigger defensive interventions.
- Persistence:** by regularly integrating time-delayed evaluation protocols into persuasion benchmarks, measuring belief decay curves at intervals from hours to months, so that the temporal reach of AI-driven persuasion can be empirically mapped to the electoral and institutional contexts in which it poses reasonable potential for harm.
- Dependence:** by instrumenting interaction logs to estimate habit-formation parameters and developing benchmarks that flag when engagement optimization reaches levels carrying reasonable potential for psychological or financial harm.
- Provenance:** by developing audit protocols that probe model training objectives and surface instances in which models were trained under covert biased objectives.